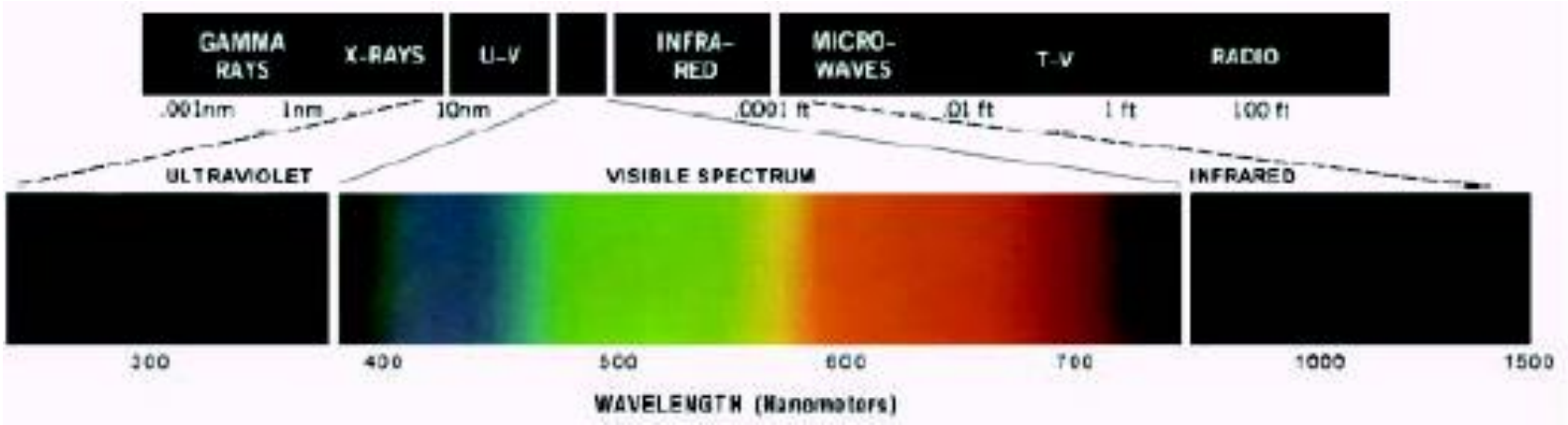
PENGOLAHAN CITRA & VISI KOMPUTER

Color Image Processing

(sumber: R.C. Gonzalez & R.E. Woods, Digital Image Processing, 4th edition)

Handayani Tjandrasa

COLOR FUNDAMENTAL



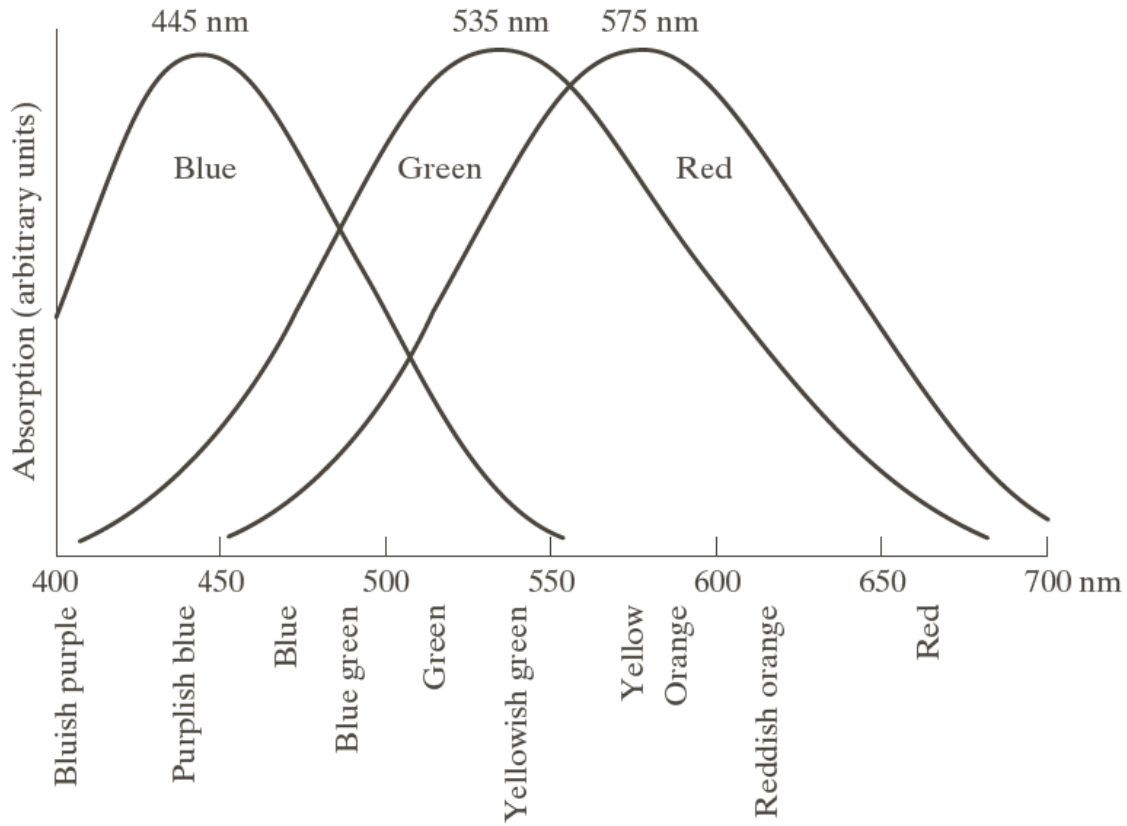
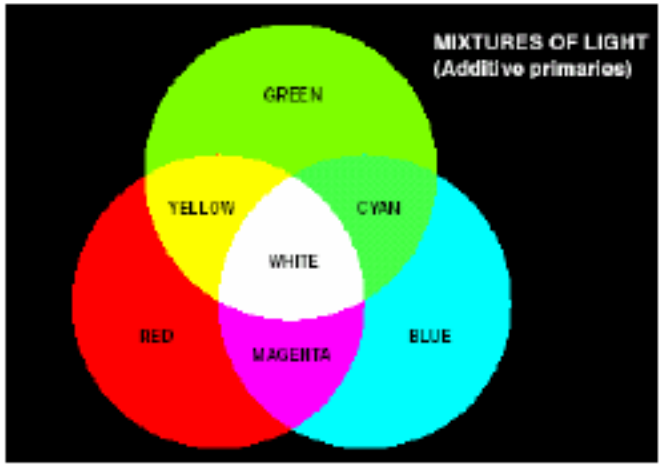


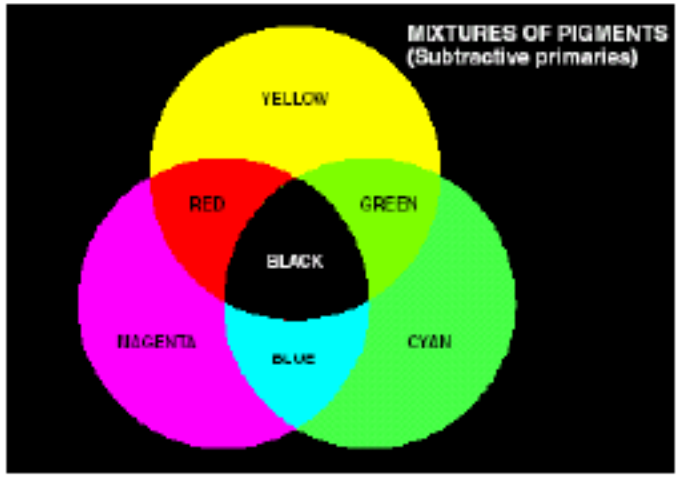
FIGURE 6.3
Absorption of light by the red, green, and blue cones in the human eye as a function of wavelength.

Primary Colors: Red(R), Green(G), Blue(B)

•*The primary colors can be mixed to generate secondary colors such as magenda (red+blue), cyan(green + blue) and yellow (red+green).*



Primary Colors



Secondary Colors

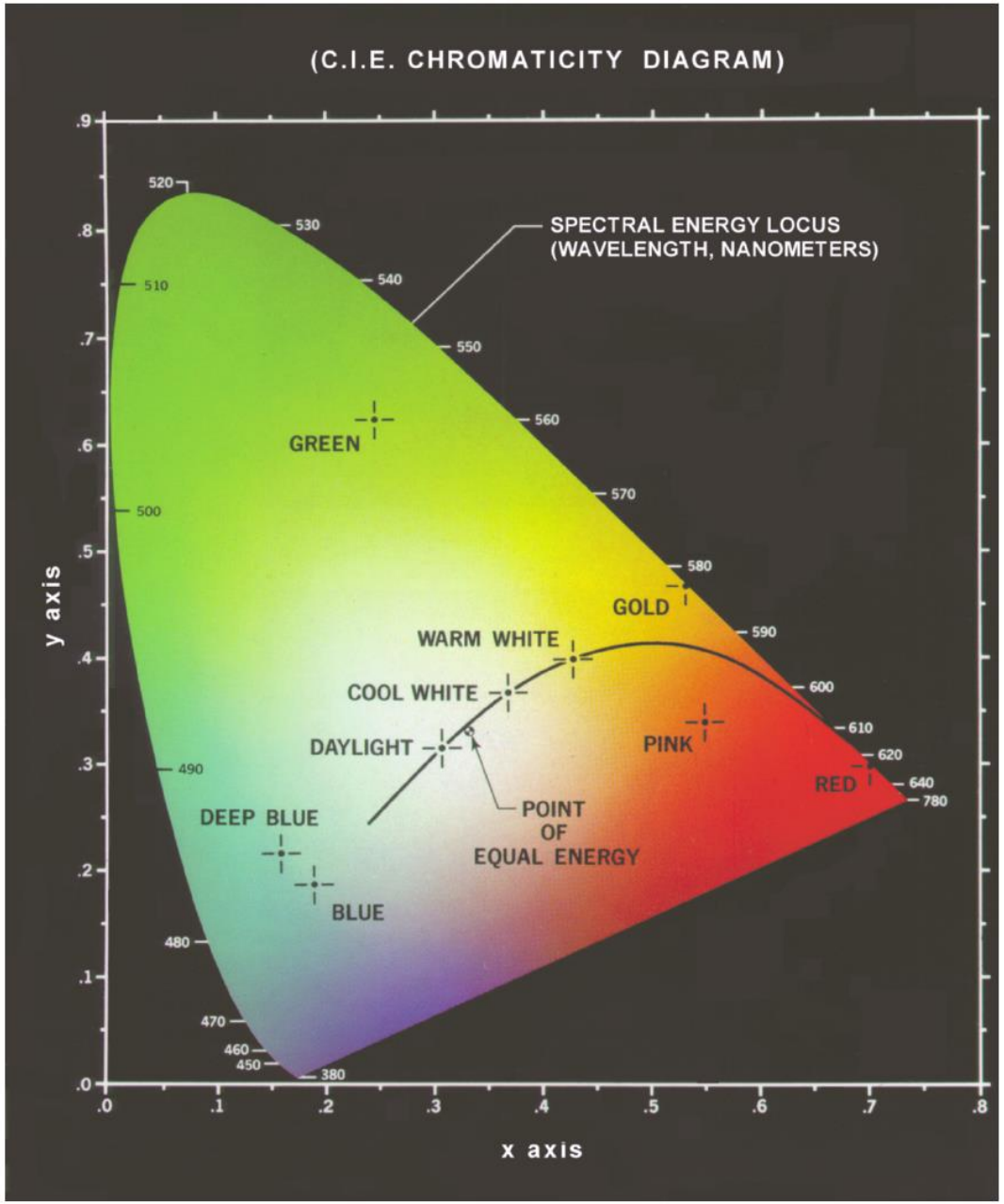
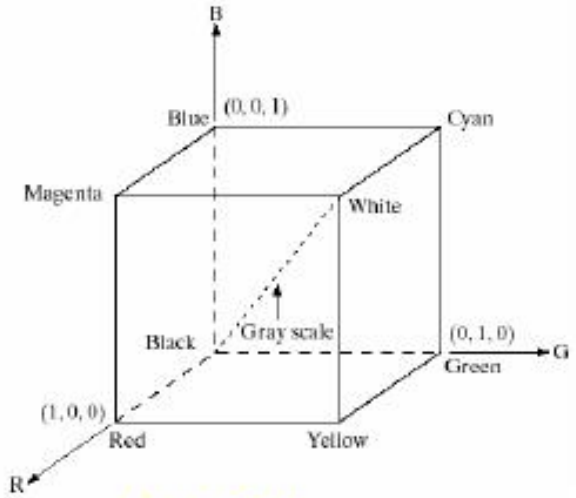
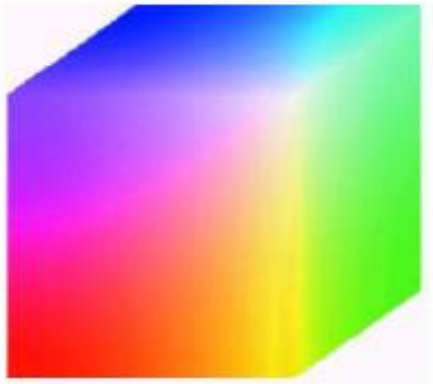


FIGURE 6.5 Chromaticity diagram. (Courtesy of the General Electric Co., Lamp Business Division.)

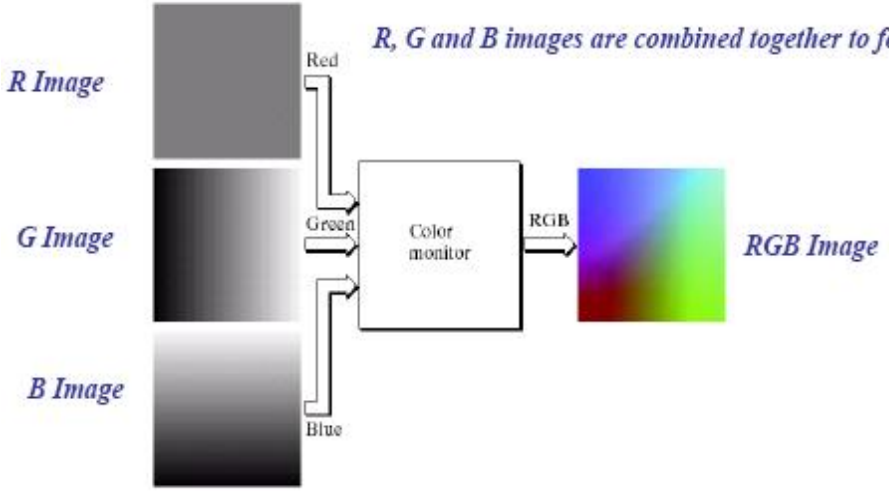
RGB color model



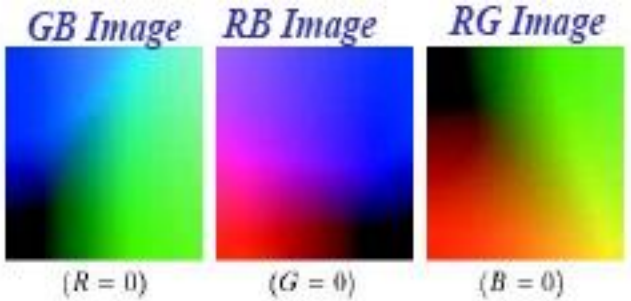
Color Cube
0,0,0=Black
1,1,1=White



24-bit Color Cube



R, G and B images are combined together to form a color image.



CMY color model:

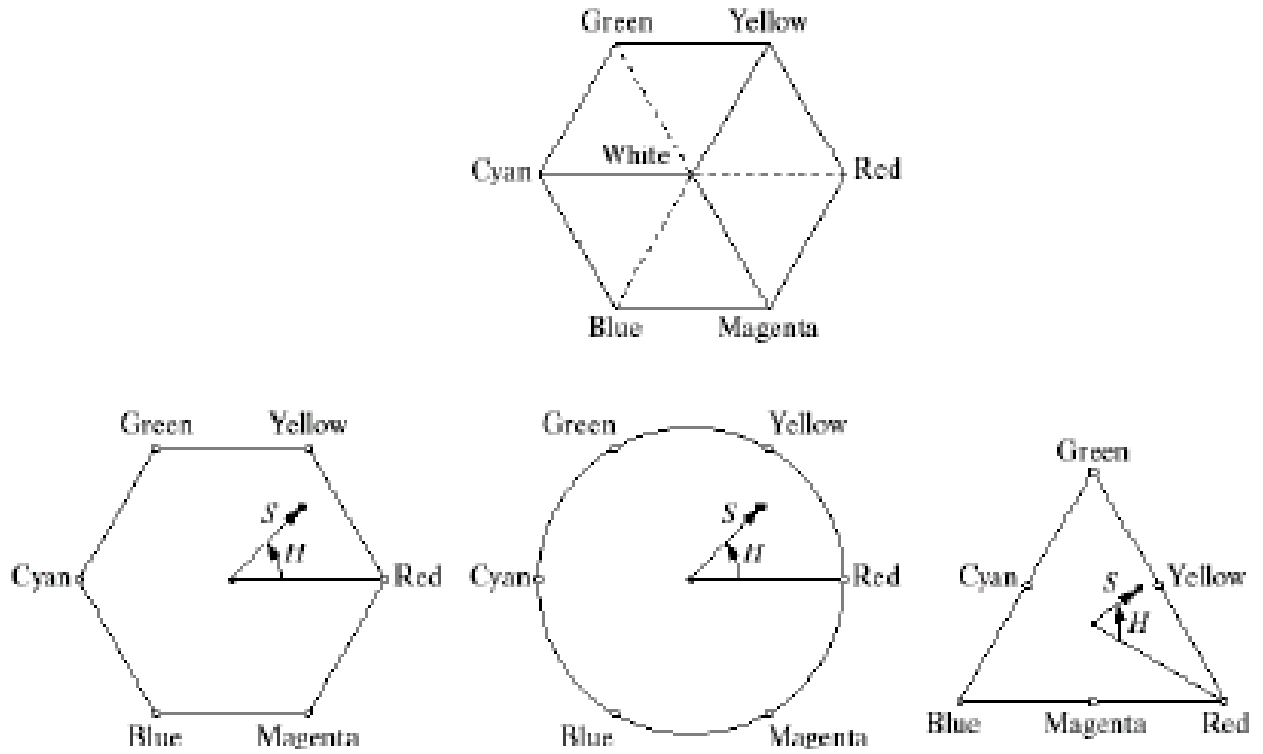
Secondary colors: Cyan, Magenta, Yellow

- *The conversion from RGB to CMY can be performed as follows:*

$$\begin{bmatrix} C \\ M \\ Y \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

- ***HSI Color Model:*** *Hue, Saturation and Intensity are three important descriptors used by human being in describing colors.*
- ***Hue*** *represents the purity of the color. (i.e. pure red, yellow, green).*
- ***Saturation*** *represents the measure of the degree to which a pure color is diluted by white light.*
- ***Intensity*** *is the gray level value of the color.*
- *Hue and Saturation represents the color carrying* ***Chrominance*** *(Chromatic) information.*
- *Intensity represents the gray-level* ***Luminance*** *(achromatic) information.*

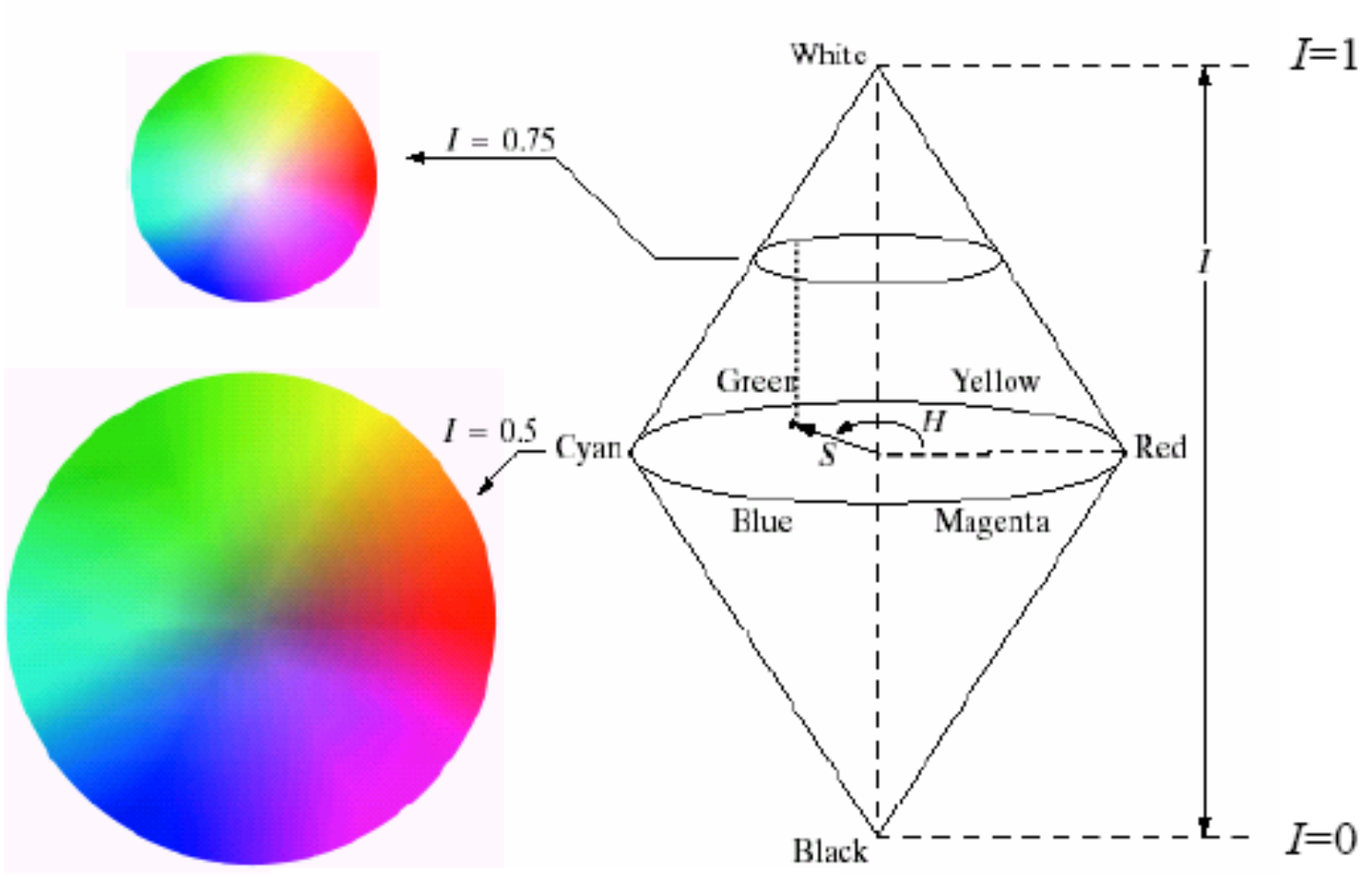
HSI color model: Hue, Saturation, Intensity



a
b c d

FIGURE 6.13 Hue and saturation in the HSI color model. The dot is an arbitrary color point. The angle from the red axis gives the hue, and the length of the vector is the saturation. The intensity of all colors in any of these planes is given by the position of the plane on the vertical intensity axis.

3D circular HSI color plane



Converting color from RGB to HSI

$$H = \begin{cases} \theta & if \quad B \le G \\ 360 - \theta & if \quad B > G \end{cases}$$

$$\theta = \cos^{-1}\left\{ \frac{\frac{1}{2}\left[(R-G)+(R-B)\right]}{\left[(R-G)^2+(R-B)(G-B)\right]^{1/2}} \right\}$$

$$S = 1 - \frac{3}{(R+G+B)}[\min(R,G,B)]$$

$$I = \frac{1}{3}(R+G+B)$$

- *The RGB values are normalized to the range [0,1].*
- *The SI values are in [0,1] and the H value can be divided by 360º to be in the same range.*

Converting from HSI to RGB

•If ($0 \leq H < 120^o$), then color is in RG Sector and then:

$$B = I(1 - S)$$

$$R = I\left[1 + \frac{S\cos H}{\cos(60^o - H)}\right]$$

$$G = 1 - (R + B)$$

•If ($120^o \leq H < 240^o$), then color is in GB Sector and then. Firstly H in this sector is:

$$H = H - 120^o$$

$$R = I(1 - S)$$

$$G = I\left[1 + \frac{S\cos H}{\cos(60^o - H)}\right]$$

$$B = 1 - (R + G)$$

•*If ($240^o \leq H < 360^o$), then color is in BR Sector and then. Firstly H in this sector is:*

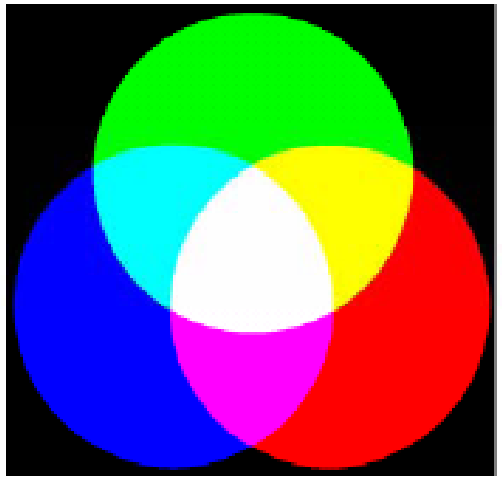
$$H = H - 240^o$$

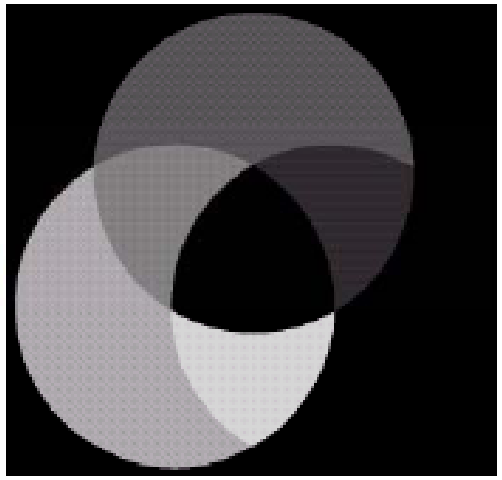
$$G = I(1 - S)$$

$$B = I\left[1 + \frac{S\cos H}{\cos(60^o - H)}\right]$$

$$R = 1 - (G + B)$$

RGB image

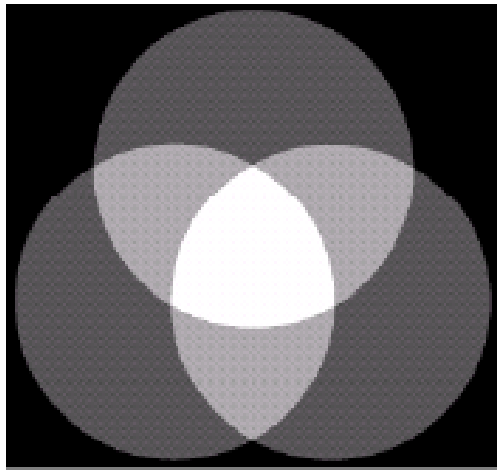




H Image

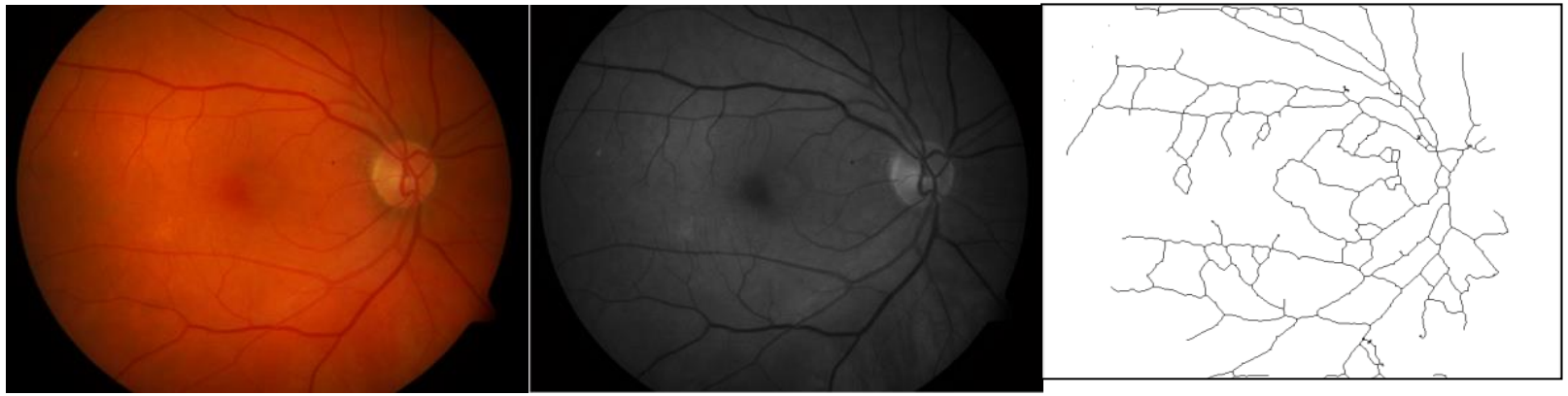
S Image



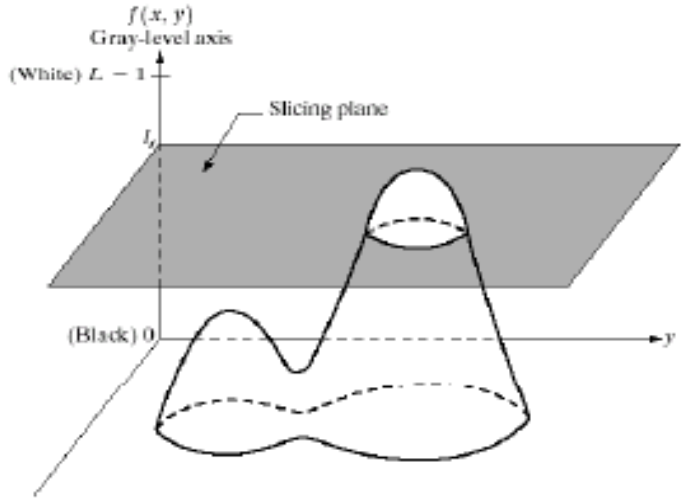


I Image

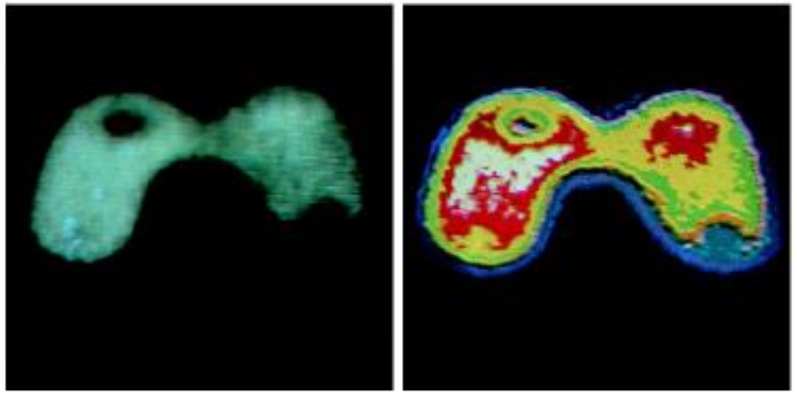
Example: Converting a color retina image into green channel and segmentation result



Pseudocolor Image Processing

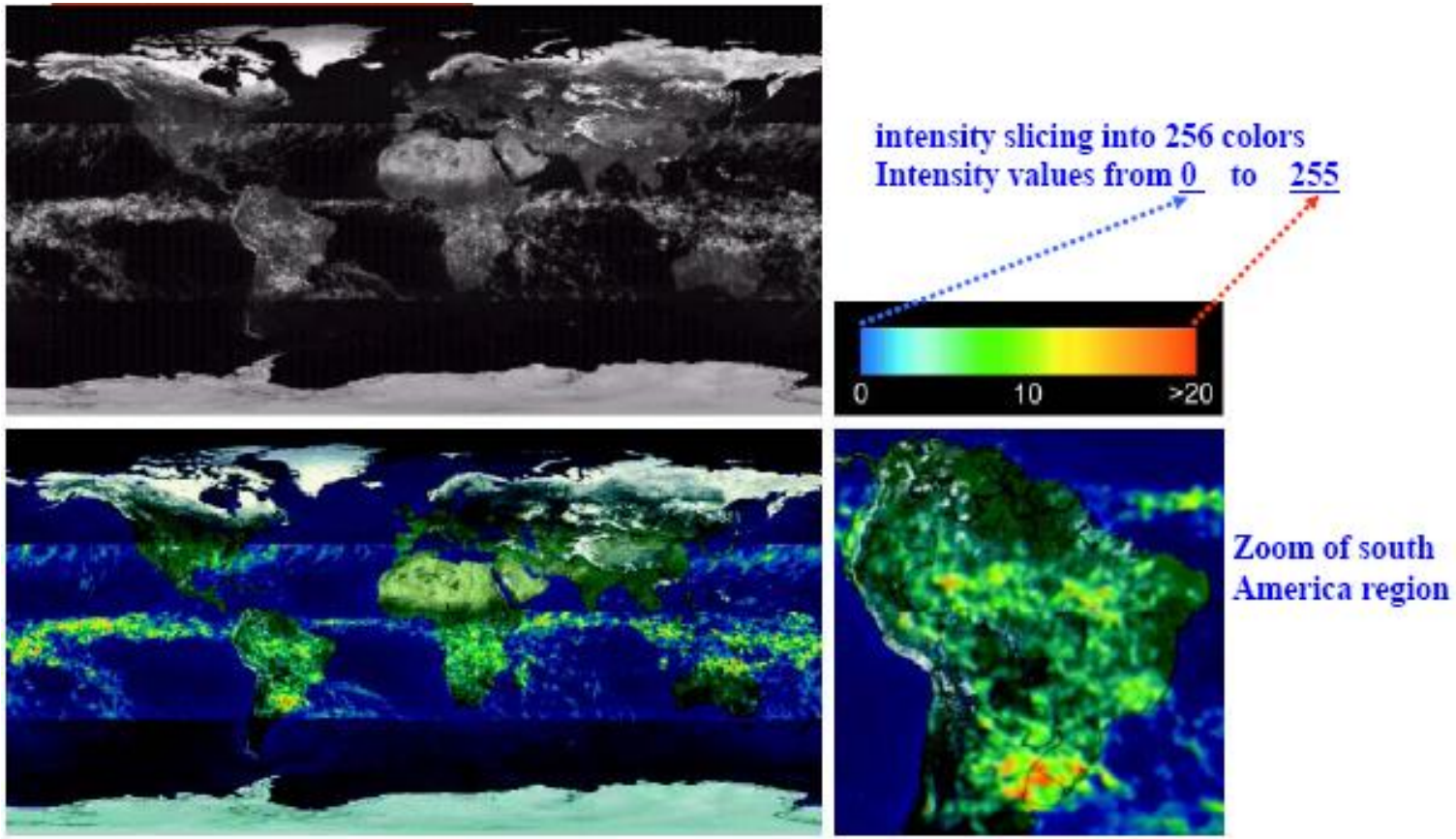


Intensity slicing

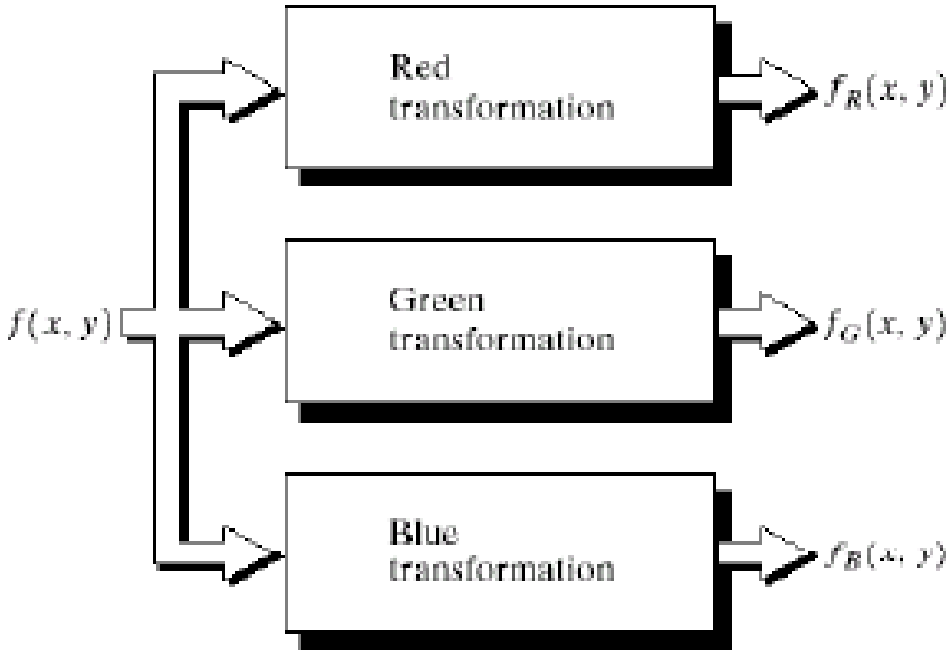


Monochrome Image

Intensity slicing into 8 colors

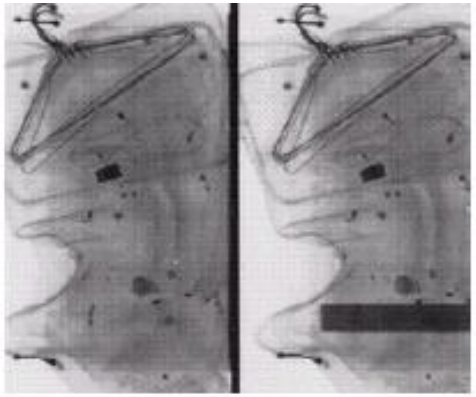

intensity slicing into 256 colors
Intensity values from 0 to 255
0
10
>20
Zoom of south America region

Gray level to color transformation

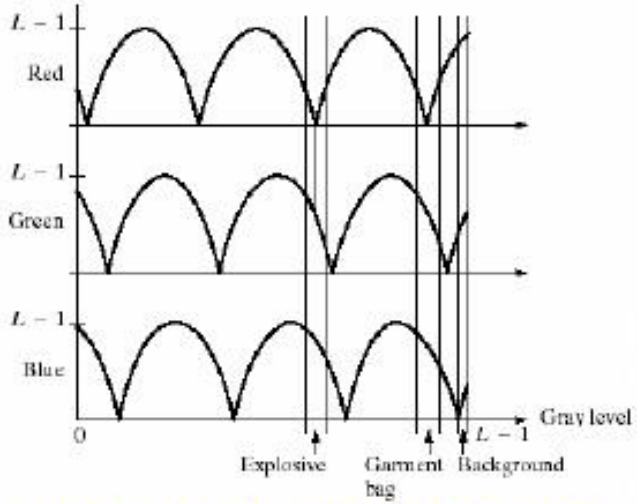


f_R, f_G and F_B are used to be inputs to an RGB monitor, producing a colored image.

Example

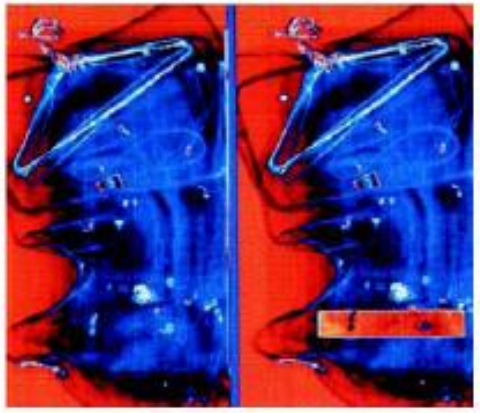


input image.



Transformation functions for R, G and B channels.

Output Pseudocolor image.



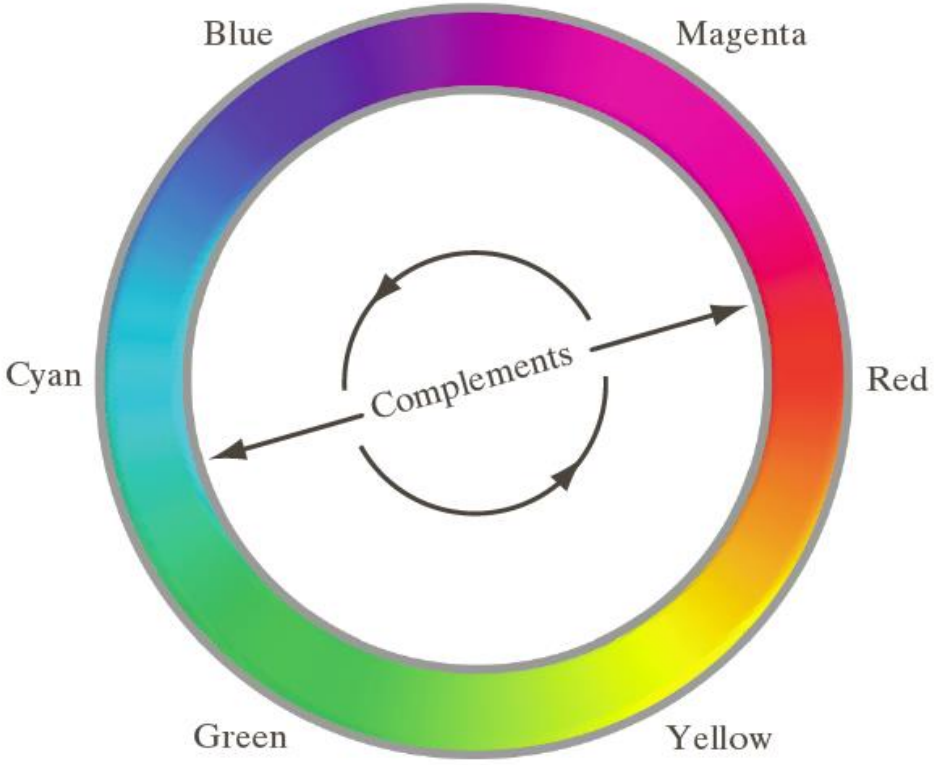
Examples: Color Image Processing

a b
c d e

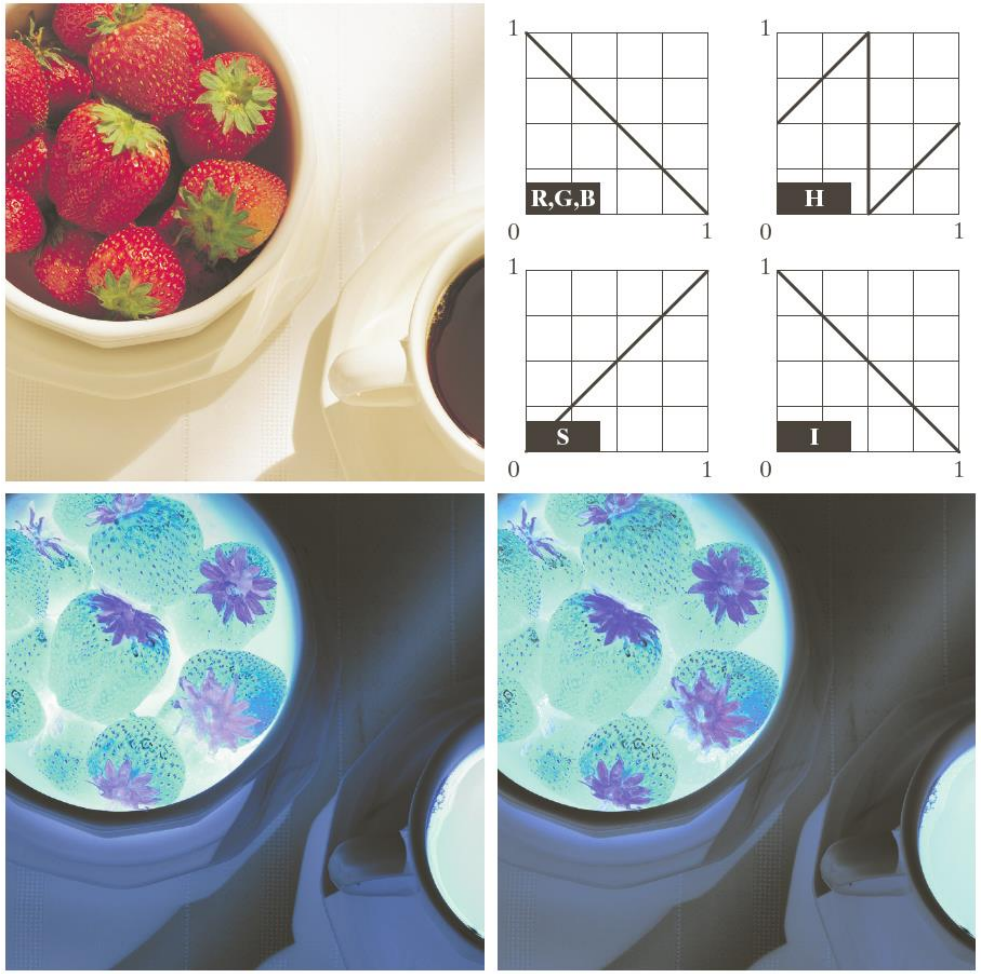
FIGURE 6.31 Adjusting the intensity of an image using color transformations. (a) Original image. (b) Result of decreasing its intensity by 30% (i.e., letting $k = 0.7$). (c)–(e) The required RGB, CMY, and HSI transformation functions. (Original image courtesy of MedData Interactive.)



Examples: Color Image Processing



Examples: Color Image Processing



a b
c d

FIGURE 6.33 Color complement transformations. (a) Original image. (b) Complement transformation functions. (c) Complement of (a) based on the RGB mapping functions. (d) An approximation of the RGB complement using HSI transformations.

Examples: Color Image Processing

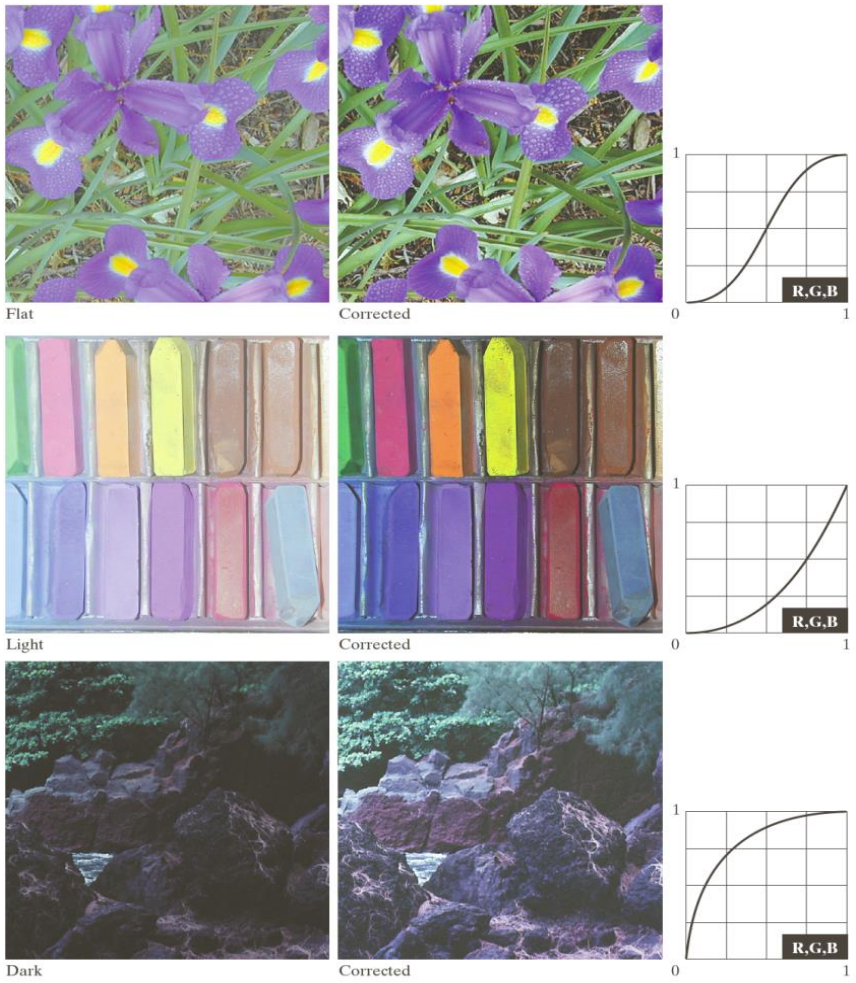


FIGURE 6.35 Tonal corrections for flat, light (high key), and dark (low key) color images. Adjusting the red, green, and blue components equally does not always alter the image hues significantly.

Examples: Color Image Processing



a b
c d

FIGURE 6.38
(a) RGB image. (b) Red component image. (c) Green component. (d) Blue component.

Examples: Color Image Processing



a b c

FIGURE 6.39 HSI components of the RGB color image in Fig. 6.38(a). (a) Hue. (b) Saturation. (c) Intensity.

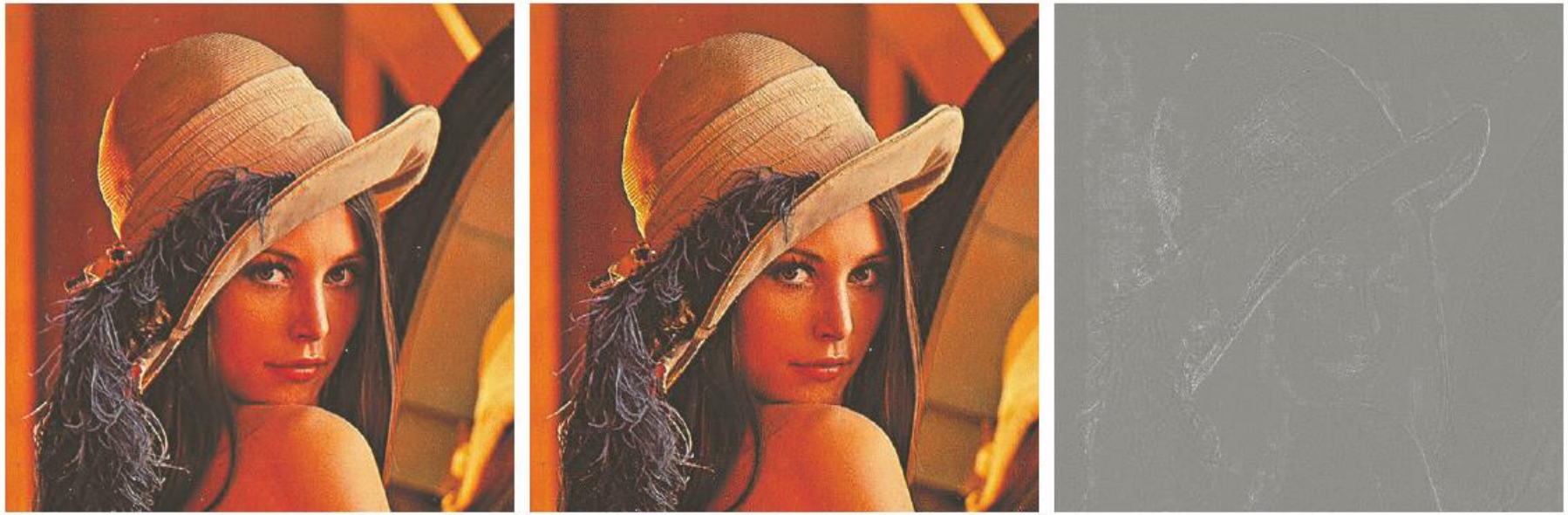
Examples: Color Image Processing



a b c

FIGURE 6.40 Image smoothing with a 5×5 averaging mask. (a) Result of processing each RGB component image. (b) Result of processing the intensity component of the HSI image and converting to RGB. (c) Difference between the two results.

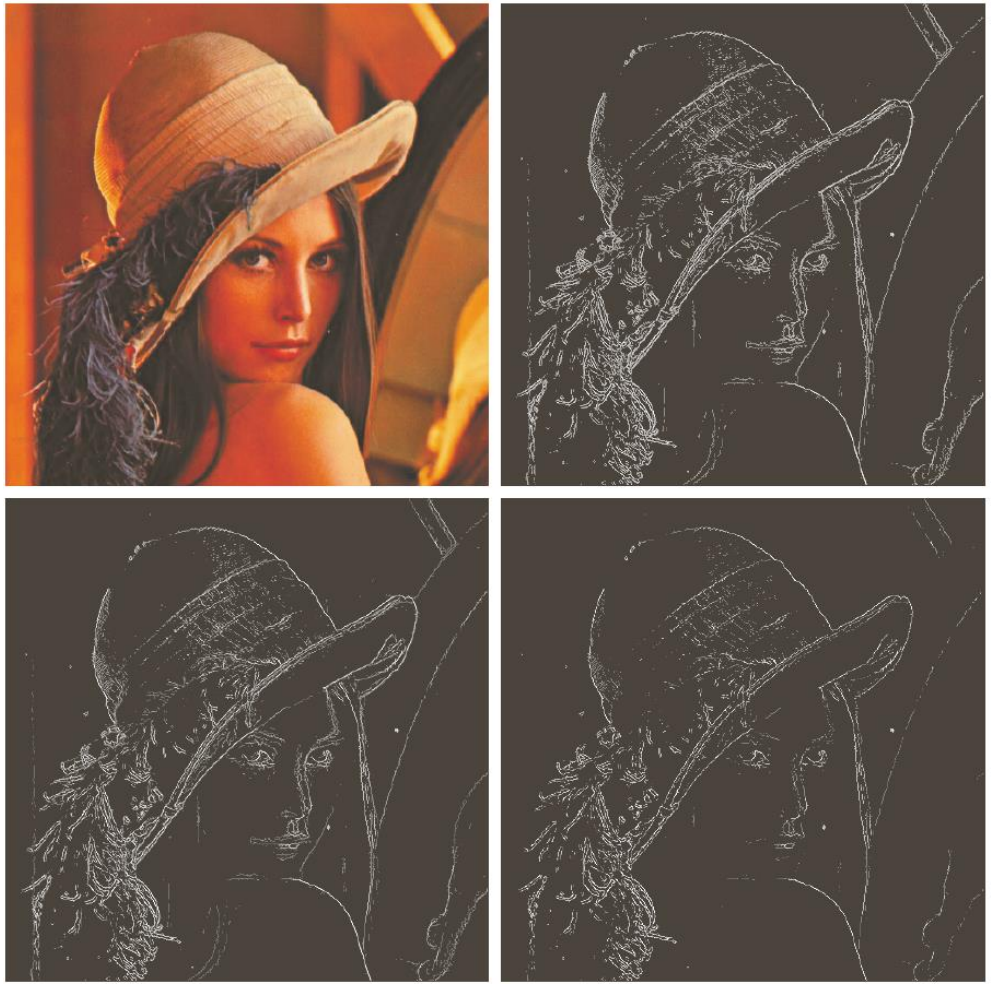
Examples: Color Image Processing



a b c

FIGURE 6.41 Image sharpening with the Laplacian. (a) Result of processing each RGB channel. (b) Result of processing the HSI intensity component and converting to RGB. (c) Difference between the two results.

Examples: Color Image Processing



a b
c d

FIGURE 6.46
(a) RGB image.
(b) Gradient computed in RGB color vector space.
(c) Gradients computed on a per-image basis and then added.
(d) Difference between (b) and (c).

Examples: Color Image Processing



a b c

FIGURE 6.47 Component gradient images of the color image in Fig. 6.46. (a) Red component, (b) green component, and (c) blue component. These three images were added and scaled to produce the image in Fig. 6.46(c).


THANK YOU